Class 1: Course Introduction

STA 220
Mai Ghannam
Summer 2026

Overview

- Welcome to STA220!
- Overview of the course and the syllabus
- Topics Covered

  - Module 1: Summarizing Data of One Variable

  - Module 1: Relationships Between Variables

Overview of the Course

What is STA220 about?

- A first course in statistics
- No prior knowledge of statistics is assumed
- **Statistics**: The process of collecting data, analyzing data, interpreting the results, and presenting conclusions.
- We focus on understanding fundamentals and how to apply statistics (as opposed to mathematical theories)

What is STA220 about?

- In this course, you'll be…
  - Learning the basics of statistical analysis, as well as the reasoning and motivations for them
    - i.e. why do we need statistics?
  - Determining appropriate uses and applications of many common statistical methods
    - i.e. what is the correct tool for my data?
  - Interpreting and using the results of these methods
    - i.e. what do my results mean?
  - Developing statistical literacy, including recognizing the importance of data in decision-making

Communication with the Teaching Team

- All questions about the course content should be posted in Piazza
  - Questions on Piazza will be answered by the course instructor.
- All questions related to course administration should be sent to sta220@course.utoronto.ca
  - Eg. accessibility accommodations during assessments, info about TA office hours
- Missed assessments: submit form
- Regrade requests: submit form
- Instructor and TA office hours will be posted on Quercus soon

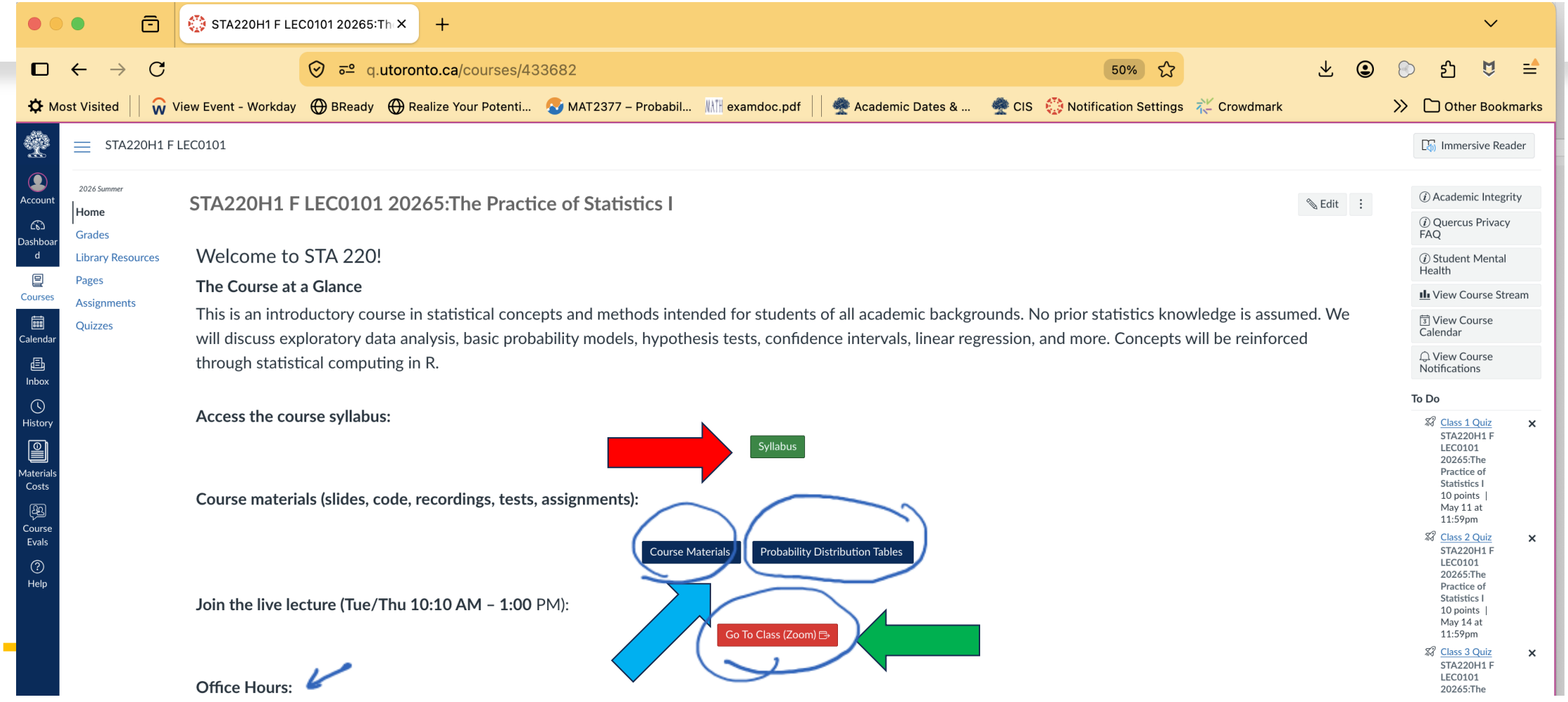
on Quercus starting May 11

Course assessment structure

- Weekly Quizzes (Online through Quercus) – 15% (best 8/10)
- **Online take-home** Test (2 hours, May 21) – 25%
- Assignment (due June 14) – 20%
- In-person final exam (TBA) – 40%
- Generative AI Usage Survey on Quizzes (Online through Quercus) – bonus 1%, to qualify you need to complete accompanying survey for 8/10 quizzes

MC + some Fill in blank

Course Site


STA220H1 F LEC0101 20265:Th
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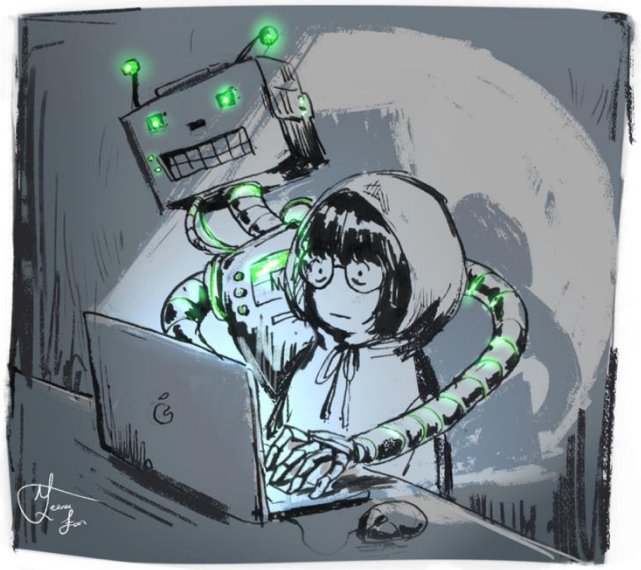
2026 Summer
Home
Grades
Library Resources
Pages
Assignments
Quizzes
STA220H1 F LEC0101 20265:The Practice of Statistics I
Edit
Welcome to STA 220!
The Course at a Glance
This is an introductory course in statistical concepts and methods intended for students of all academic backgrounds. No prior statistics knowledge is assumed. We will discuss exploratory data analysis, basic probability models, hypothesis tests, confidence intervals, linear regression, and more. Concepts will be reinforced through statistical computing in R.
Access the course syllabus:
Syllabus
Course materials (slides, code, recordings, tests, assignments):
Course Materials
Probability Distribution Tables
Join the live lecture (Tue/Thu 10:10 AM – 1:00 PM):
Go To Class (Zoom)
Office Hours:
Academic Integrity
Quercus Privacy FAQ
Student Mental Health
View Course Stream
View Course Calendar
View Course Notifications
To Do
Class 1 Quiz STA220H1 F LEC0101 20265:The Practice of Statistics I 10 points | May 11 at 11:59pm
Class 2 Quiz STA220H1 F LEC0101 20265:The Practice of Statistics I 10 points | May 14 at 11:59pm
Class 3 Quiz STA220H1 F LEC0101 20265:The

Use of Generative AI in this course

- From the syllabus:
  - *"Students are encouraged to make use of technology, including generative artificial intelligence tools (e.g. ChatGPT), to contribute to their understanding of course materials."*
- But you are ultimately accountable for the work you submit
  - Beware of false or factually incorrect information
  - There can be different approaches to solve a problem in statistics; the idea is to use concepts taught in the course
- Want to avoid over-reliance on AI
- Main goal of course is to learn and develop critical thinking skills for making sense of data



Use of Generative AI in this course: specifics

- Following each quiz, there is an optional survey (for bonus marks) about your use of AI on the quiz (did you use it, what prompts did you use)
- For the assignment: need to document your use of AI in detail in an appendix (more information on this later)
- Students may **NOT** use AI tools for the completion of, or to support the completion of Term Test or Final Exam!

What is Statistics?

What is data?

- **Data** (plural of datum) : a *collection of numbers, characters, images, or other items that provide information about something.*
  - In other words, it is anything meaningful that we can 'measure' from an object/person/thing
- Super vague definition:
  - if data is anything, what is *good/meaningful* data?
  - once we have data, how do we make any sense of it?

What is Statistics?

- *Statistics: the science of learning from data*
  - involves the collection, organization, and interpretation of data, but also much more
- Nearly all fields use statistics: biology, chemistry, medicine, environment, law, politics, etc.
- The goal is the use data to infer facts, make predictions and decisions

Why Statistics?

- The reason for using statistics is that we never get the whole picture.
- The data we have to work with is always some smaller portion of all people/objects/things we could be taking measurements on.
- Studies used to generalize the results from the data collected to a larger population
  - Quality of the data impacts the quality of the statistical results.
  - Statistics helps define how to collect good data

But it is always important to be critical!

Gluten Free Diets Example

- Celiac disease affects ~1% of Canadians, and ~6% have physician-diagnosed gluten sensitivity.
- Despite this, gluten-free diets became a major trend among people without gluten sensitivity.

Gluten Free Diets Example: Research Question

- Is a gluten-free diet beneficial for people without gluten sensitivity?
- A randomized double-blind study investigated whether gluten affects gastrointestinal health in healthy individuals.

Gluten Free Diets Example: Study Design

- 28 healthy participants followed a gluten-free diet for 2 weeks.
- Half consumed gluten flour and half consumed gluten-free flour without knowing which they received (double blind randomized trial)

Gluten Free Diets Example: Results

- Gastrointestinal health was statistically the same in both groups.
- Gluten did not induce GI symptoms in healthy volunteers.

Gluten Free Diets Example: Questions This Raises

- How do we design and evaluate experiments like this?
- How do we interpret statistical evidence and sample sizes?

Gluten Free Diets Example: Connection to the course

- Statistics helps answer questions about experiments, data, and uncertainty.
- This course will introduce tools to evaluate studies like this one.

Summarizing Data: One Variable

Five-Number Summary

Suppose we have data and we order it in increasing order.

1) **Minimum:** smallest observation

2) **Maximum:** largest observation

3) **Median:** observation halfway from smallest to largest observation

4) **First Quartile:** point that is a quarter of the way along the sorted data

5) **Third Quartile:** point that is three quarters of the way along the sorted data

bulk 50% of your data

Five-Number Summary
Example 1

79, 68, 88, 69, 90, 74, 87, 93, 76

① sort in increasing order:

68, 69, 74, 76, 79, 87, 88, 90, 93

5-num summary

① Min = 68, ② max = 93

③ Median = 79

④ 1st quartile = 74

⑤ 3rd Quartile = 88

Five-Number Summary
Example 2

78, 93, 68, 84, 90, 74

$\Rightarrow$ 68, 74, 78, 84, 90, 93

① Min = 68

② Max = 93

③ Median = $\frac{78+84}{2} = 81$ (2nd Quartile)

④ 1st Quartile = 74

⑤ 3rd Quartile = 90

Five-Number Summary
Formula

percent

sample size

$$\boxed{1 + p(n - 1)}$$

Formula to get position of p% percentile of sorted observations.

1st Quartile $\Rightarrow$ $p = 0.25$

$\Rightarrow$ 1st Quartile position $= 1 + 0.25(n - 1)$

Five-Number Summary
Formula

(*) 3rd Quartile:

use $p=0.75$

(*) Median:

use $p=0.5$

Let's use the formula for Example 2:

78, 93, 68, 84, 90, 74 → $n=6$

68, 74, 78, 84, 90, 93

2.25 position

1st Quartile position $= \underbrace{1+0.25(6-1)}$ $= 1+0.25(5) = 1+1.25$
$= 2.25$

2nd position

get quarter way past 2nd

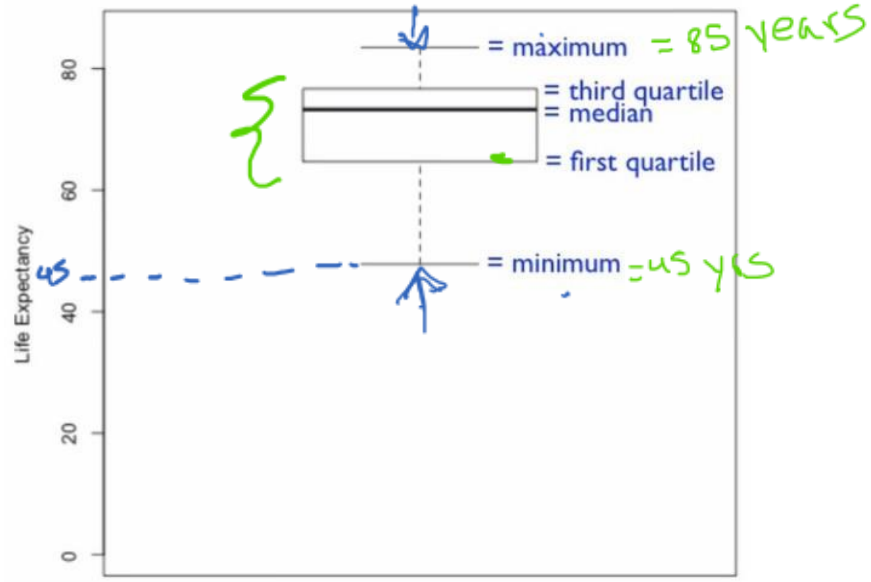
1st Quartile $= 74 + 0.25(78-74) = \underline{75}$

2nd position

quarter of the way more

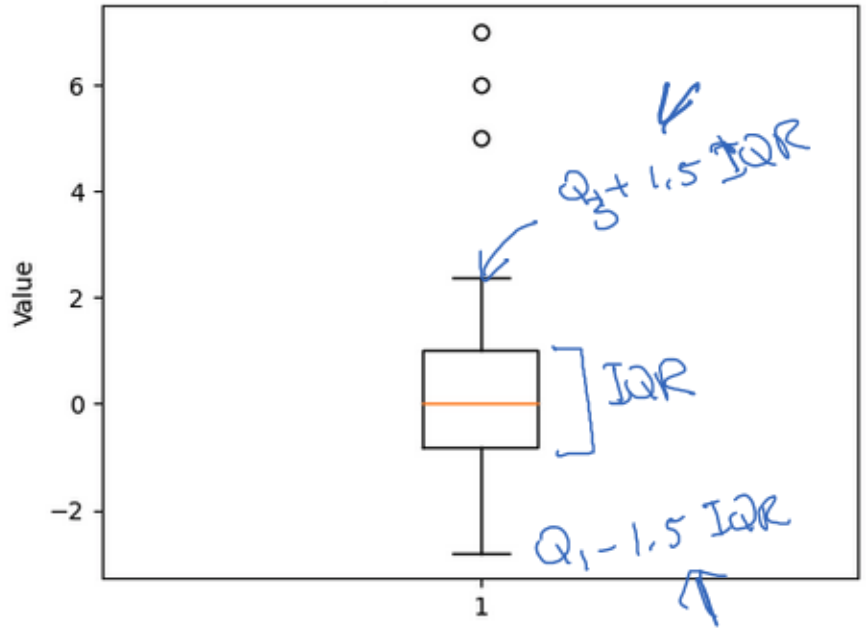
Boxplots
Unmodified Boxplot

- The five number summary can be visualized using a boxplot (unmodified)



Boxplots
Modified Boxplot

- Modified boxplots help identify extreme or unusual values.
- They use the interquartile range (IQR) to determine outliers.
- IQR measures the spread of the middle 50% of the data.



Boxplots
Interquartile Range and Fences

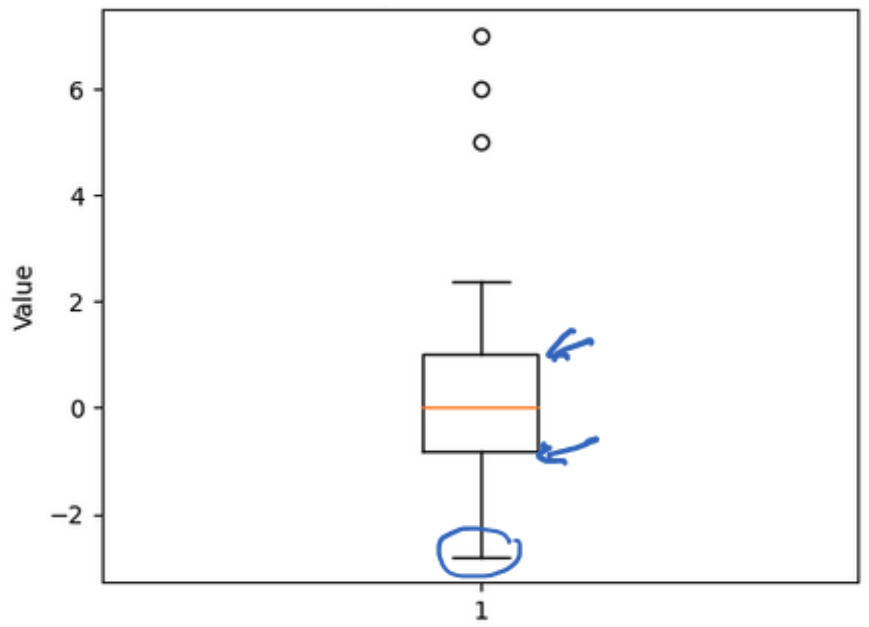
- IQR measures the spread of the middle 50% of the data.

3rd Quartile

- $\boldsymbol{IQR = Q_3 - Q_1}$

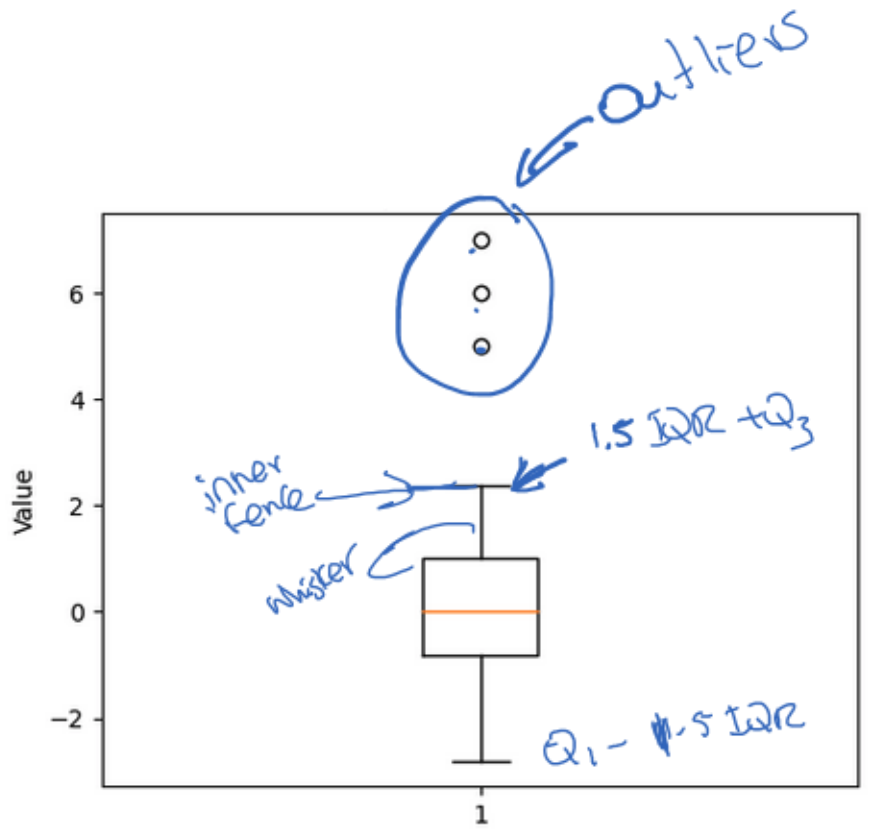
1st Quartile

  - Inner Fences:
    - Lower:
      $$\boldsymbol{Q_1 - 1.5 \times IQR}$$
    - Upper:
      $$\boldsymbol{Q_3 + 1.5 \times IQR}$$

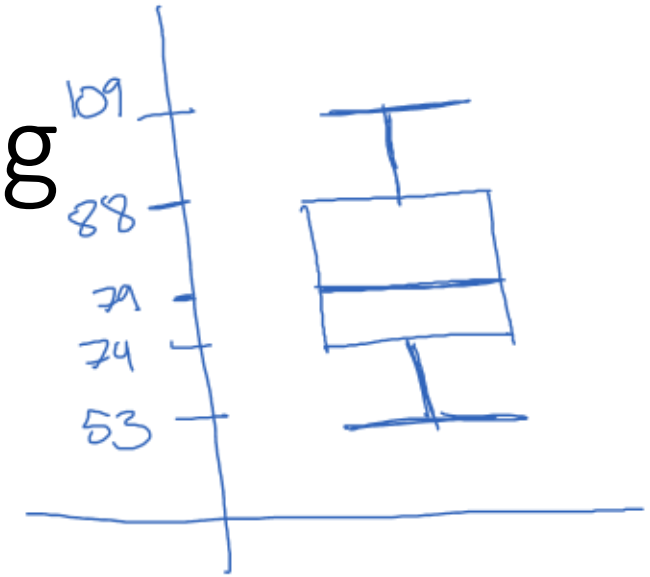


Boxplots
Whiskers and Outliers

- Whiskers extend to the most extreme data values within the inner fences.
- Values outside the fences are plotted individually as outliers.



IQR and Fence Calculation Using Previous Example:



79, 68, 88, 69, 90, 74, 87, 93, 76

68, 69, 74, 76, 79, 87, 88, 90, 93

1st Quartile = 74

Median = 79

3rd Quartile = 88

Min = 68

Max = 93

IQR = $Q_3 - Q_1$ = 88 − 74 = 14

Lower Fence = $Q_1 - 1.5\,IQR$

= 74 − 1.5 (14) = 53

Upper Fence = $Q_3 + 1.5\,(14)$

= 88 + 1.5 (14)

= 109

Centre and Spread

- **Two Key Ways to Summarize Data**

- **1.** ***Where is the centre?***
- Median (50th percentile) not affected by outliers
- Mean (arithmetic average) is very affected by outlier
- $k\%$ trimmed mean (average after removing extreme values)

- **2.** ***How spread out is the data?***

- Range (max – min)
- Interquartile Range (IQR = Q3 – Q1)
- Variance and standard deviation

Measures of Centre

Measures of Centre

Median

- Middle value of the ordered data

Mean

- Average of all values
- Sensitive to outliers

k% Trimmed Mean sol^n

- Remove lowest and highest $k\%$ of values
- Compute the mean of what remains
- Reduces the effect of outliers

Measures of Centre: k% Trimmed Mean

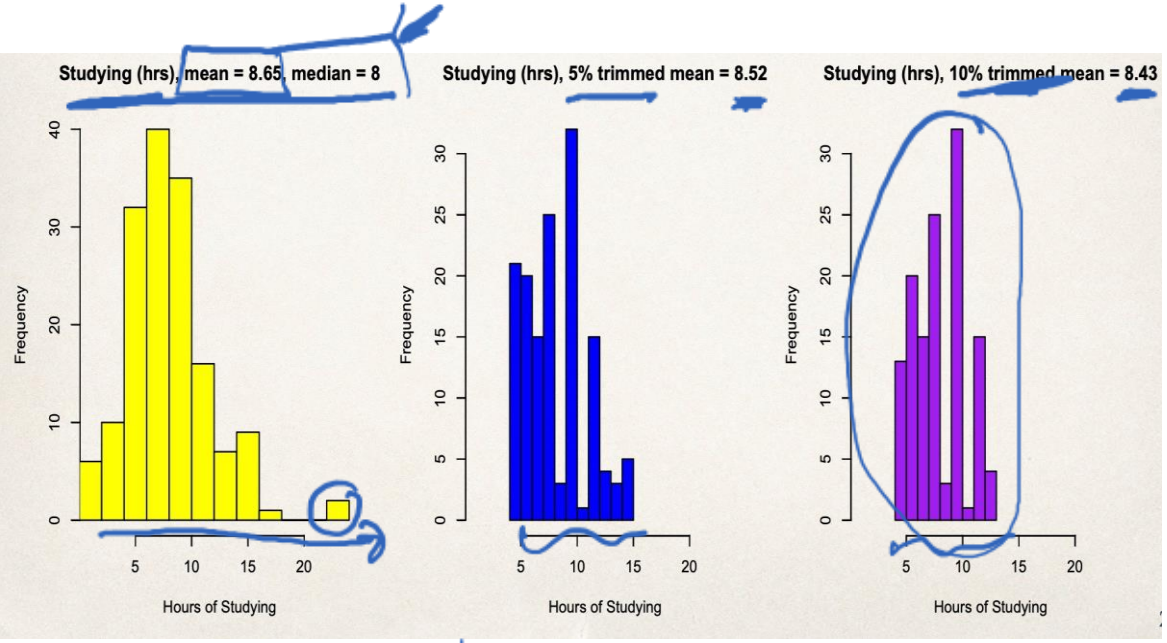
The more trimming you do → the closer you get to the median

Always round down!

Example (n=158):

- 5% trimmed → remove 7 values from each end
- 10% trimmed → remove 15 values from each end

Trimming reduces the effect of extreme values (outliers)



0.05(158) = 7 values

Remove 7 highest + 7 lowest measurements.

Remove 14

Measures of Centre Using Previous Example:

79, 68, 88, 69, 90, 74, 87, 93, 76

~~68~~, 69, 74, 76, 79, 87, 88, 90, ~~93~~

① Median = 79

② Mean = $\frac{79+68+88+69+\cdots+76}{9} = 80.44$

③ 20% trimmed Mean ⇒ $0.2(9) = 1.8 \longrightarrow$ 1 value removed above + 1 value removed below

$\frac{69+74+76+79+87+88+90}{7}$

Measures of Spread/Variability

There are several statistics we can use to describe how spread out data are:

1. Range

$$\text{Range} = Maximum - Minimum$$

- Uses only extreme values
- Very sensitive to outliers

2. Interquartile Range (IQR)

$$\boxed{\text{IQR} = Q_3 - Q_1}$$

- Spread of the **middle 50%**
- Resistant to outliers

Measures of Spread/Variability

There are several statistics we can use to describe how spread out data are:

3. Variance

$$\boxed{s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- Measures average squared deviation from the mean
- Uses all data values
- Sensitive to outliers

4. Standard Deviation

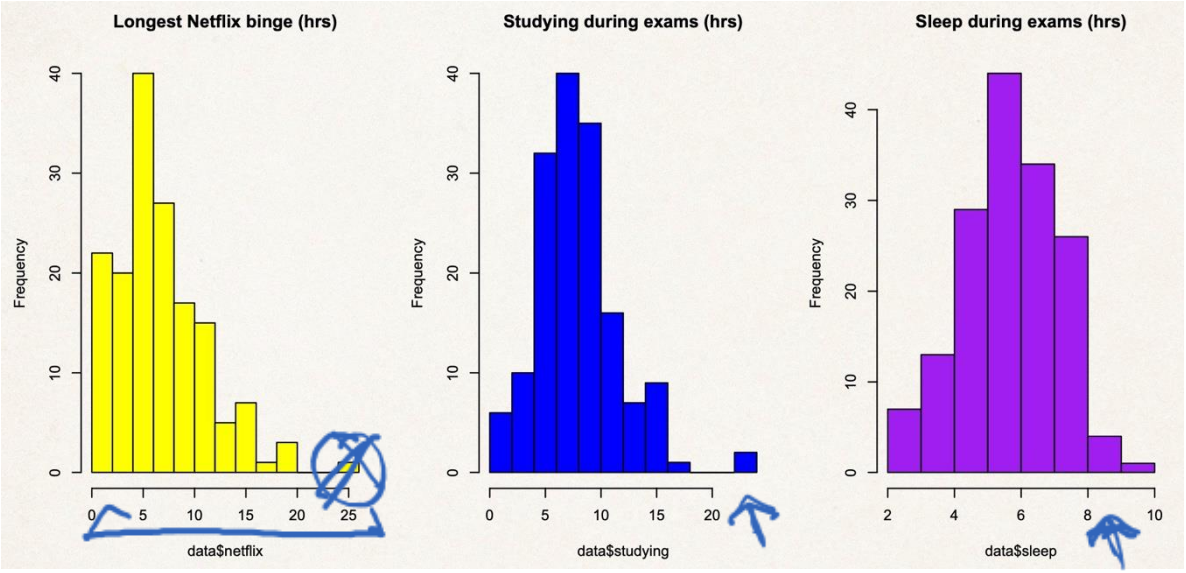
$$s = \boxed{\sqrt{s^2}}$$

$s = \sqrt{\text{Variance}}$

- Typical distance from the mean
- Same units as the data
- Most commonly used measure of spread

Range of the data

- Range will give you the whole interval where the data sits (usually a very large number)

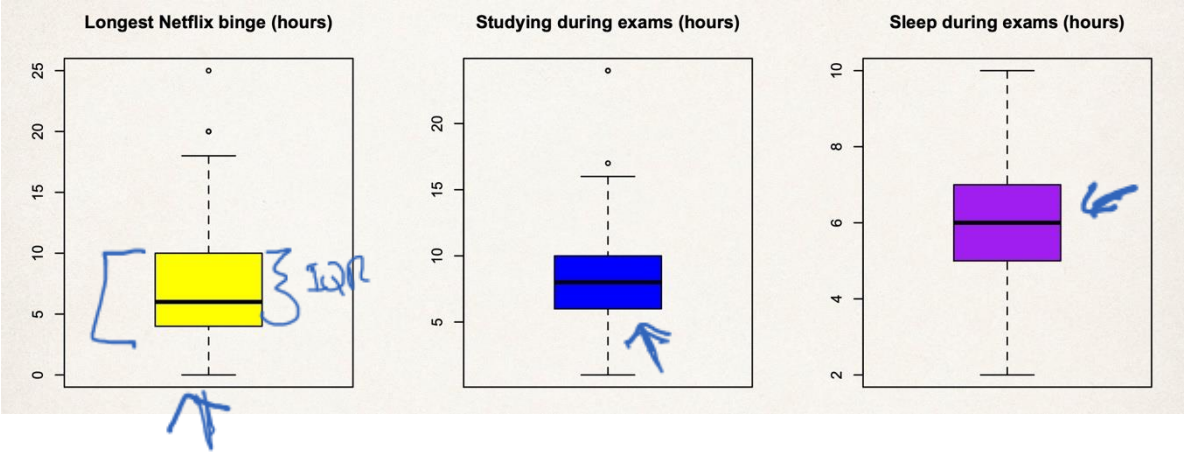


Range = 25 − 0 = 25

w/o outliers = 20 − 0 = 20

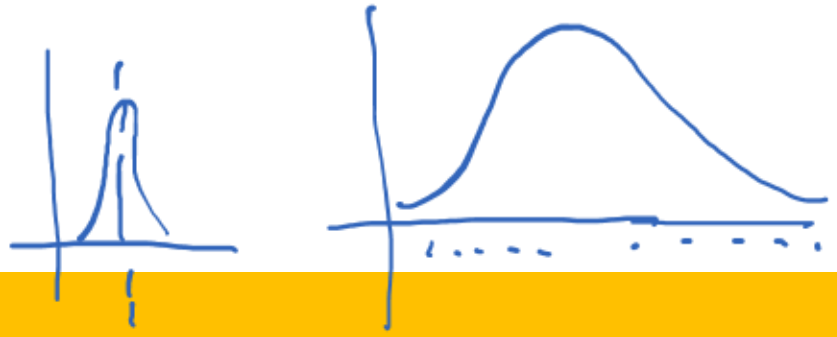
Interquartile Range (IQR)

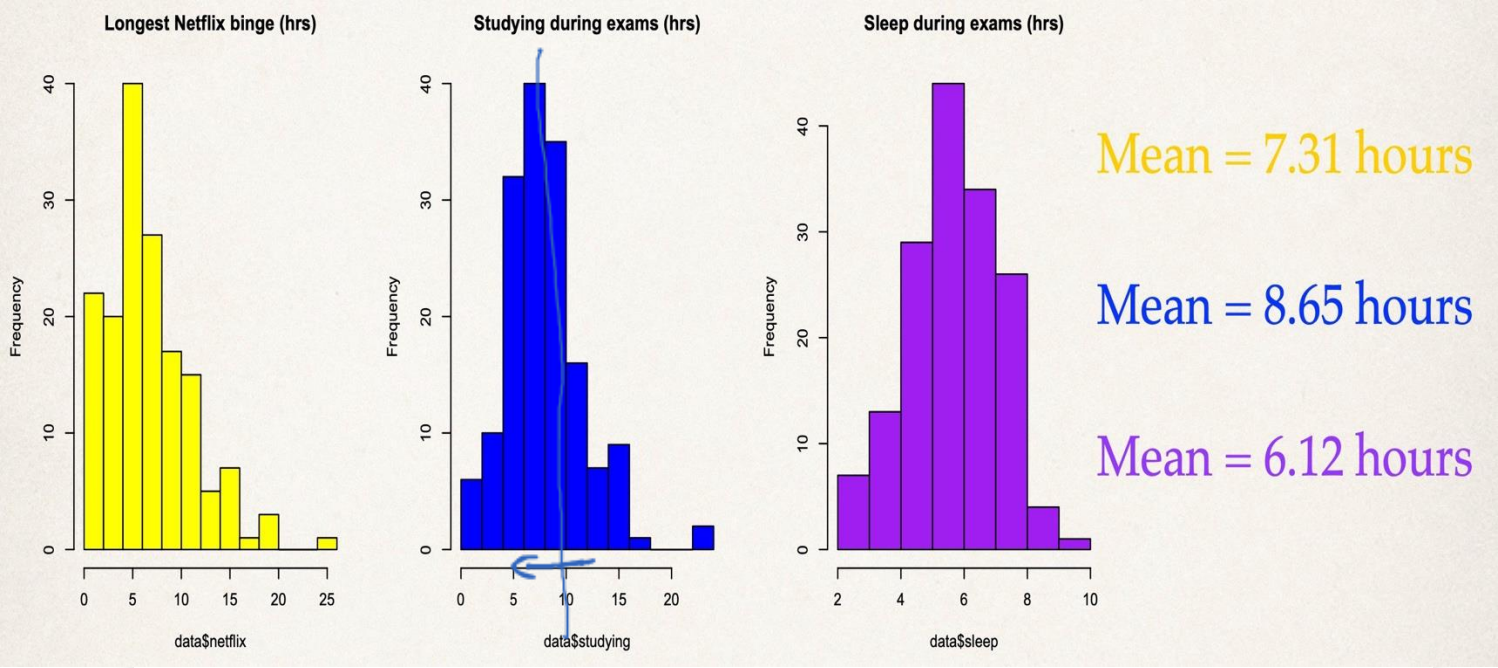
- IQR gives you the spread of the middle 50% of the data



Standard Deviation and Variance

- The standard deviation (roughly) measures the average distance between each point and the mean





Measure of Spread Using Previous Example:

1st measurement $\Rightarrow x_1 = 79$

2nd measurement $\Rightarrow x_2 = 68$

79, 68, 88, 69, 90, 74, 87, 93, 76

x_i is the ith measurement

$\bar{x} \Rightarrow$ "x bar" $\Rightarrow$ notation for the mean

① Range $= 93 - 68 = 25$

② IQR $= Q_3 - Q_1 = 88 - 74 = 14$

③ variance $= \frac{1}{n-1} \sum (x_i - \bar{x})^2 = \frac{1}{9-1}$

n: 9

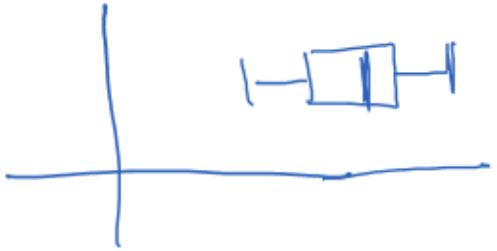
x_i: i^{th} observation

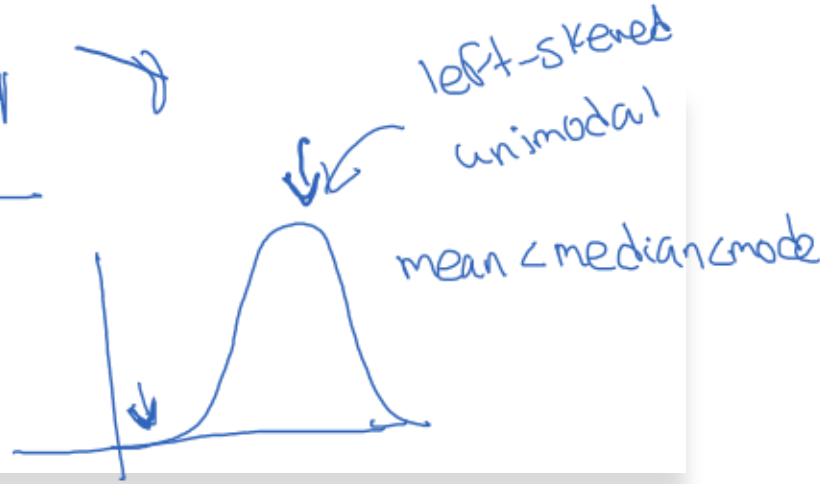
$\left((79 - 80.44)^2 + (68 - 80.44)^2 + (88 - 80.44)^2 + (69 - 80.44)^2 + \dots + (76 - 80.44)^2 \right)$

$= 87.28$

④ $s = \sqrt{\text{variance}} = \sqrt{87.28} = 9.34$

Centre, Spread and …



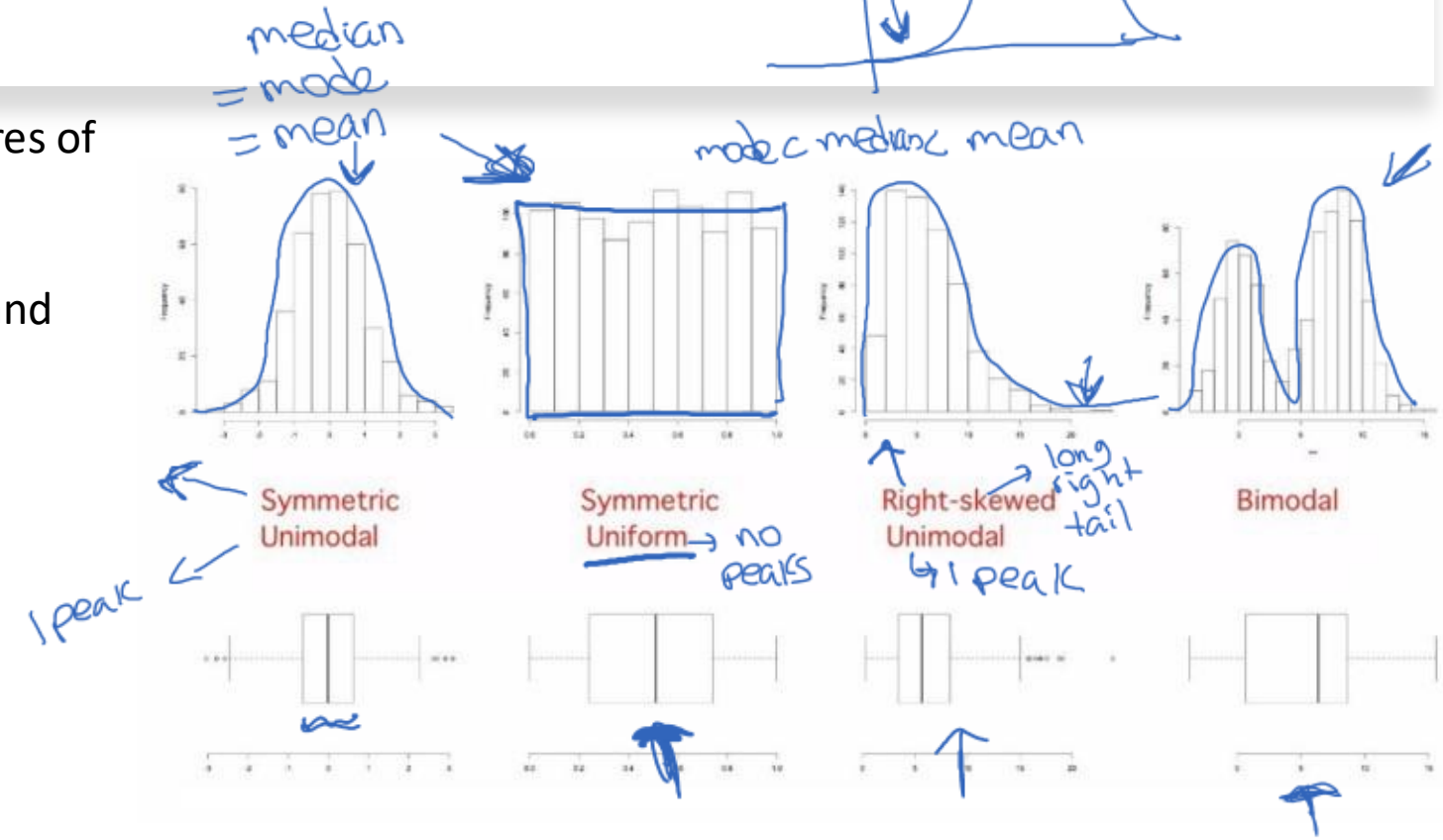


Centre and spread summarize key features of the data.

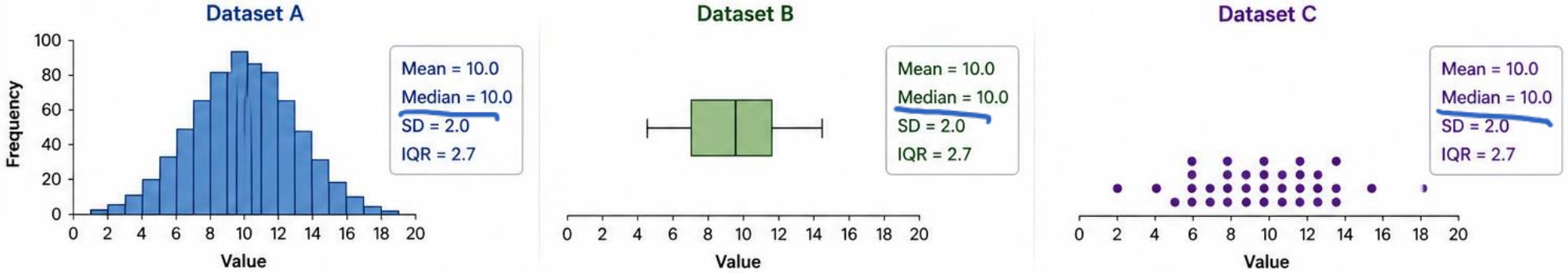
But two datasets with the same centre and spread can look very different.

The missing piece is the **shape of the distribution**.

mode = ~~#~~ value occuring the most in data

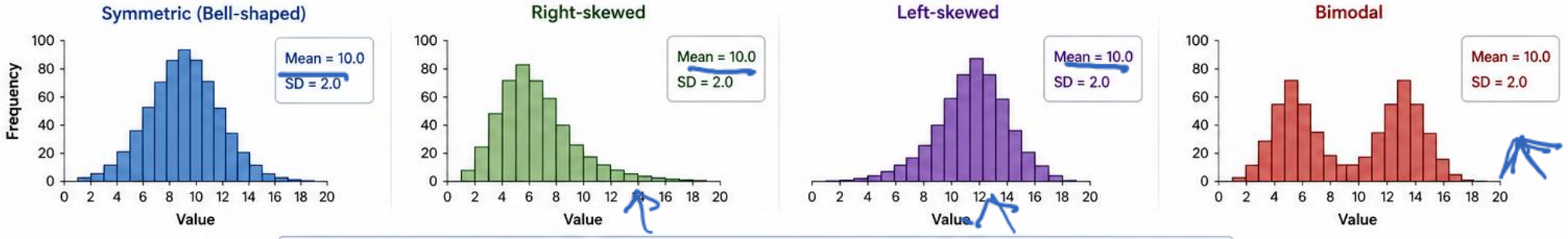


1. Centre and spread summarize key features of the data.



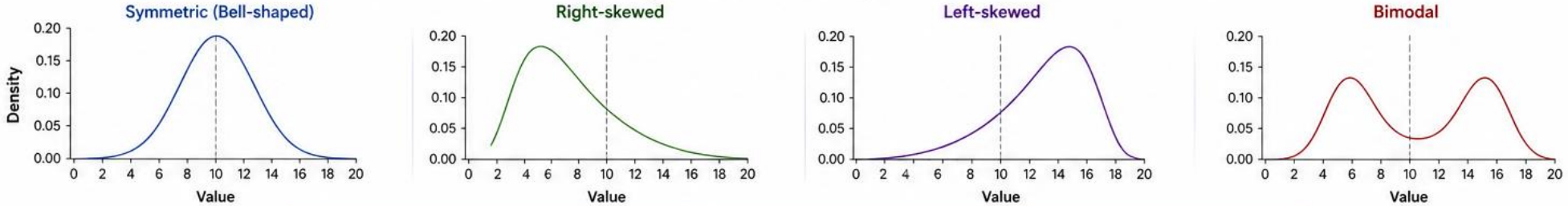
All three datasets have the same centre (mean = 10) and spread (SD = 2, IQR = 2.7).

2. But two datasets with the same centre and spread can look very different.



Each dataset has the same mean (10) and standard deviation (2), but the distributions are very different.

3. The missing piece is the shape of the distribution.



Shape describes how the data are distributed around the centre — it's the missing piece that tells the full story.

Z-Scores

Z-scores allow us to compare observations from different datasets.

They standardize values so we can make **fair comparisons (“apples to apples”)**, even when datasets have different means and standard deviations.

Z-Scores

Example

You received a grade of **80%** on a test where:

Class mean = 70%

Standard deviation = 5%

Your friend also scored **80%**, but in a different section where:

Class mean = 76%

Standard deviation = 2%

Question

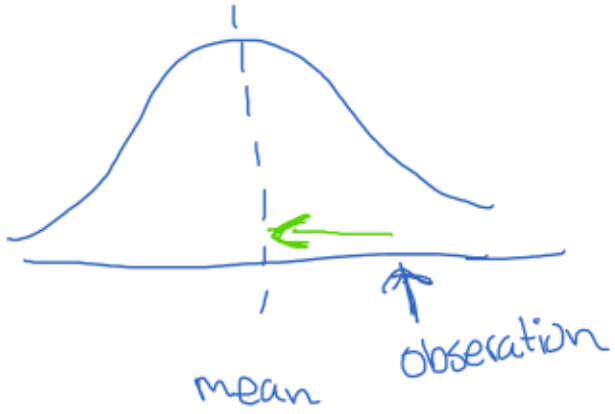
Who performed better, relative to their class?

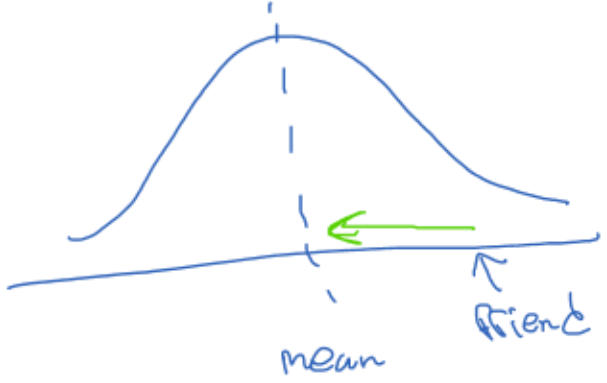
Z-Scores

Idea

To answer this fairly, we compare how far each score is from its class mean, in units of standard deviation.

Z-Scores





Comparing Scores with Z-Scores

Z-score formula:

$$z = \frac{value - mean}{standard\ deviation}$$

Your score:

$$z_{you} = \frac{your\ grade - class\ mean}{SD} = \frac{80 - 70}{5} = \frac{10}{5} = 2$$

my grade

class mean (percent)

sd (percent)

Your friend's score:

$$z_{friend} = \frac{friend's\ grade - class\ mean}{SD} = \frac{80 - 76}{2} = \frac{4}{2} = 2$$

yes we scored the same

Categorical Variables

There are two broad categories of data:

Quantitative (Numeric) Data
•Values are **numerical** and typically measured
•Examples: age, height, blood pressure

Qualitative (Categorical) Data
•Values represent **categories or labels**
•Examples: letter grades, true/false, favourite colour

Why This Matters
•The way we **visualize** and **summarize** data depends on its type
•Measures like **mean, median, variance, range**
•Graphs like **histograms** and **boxplots**
→ These apply only to **quantitative data**

Categorical Variables

Distribution of a Variable

- The **distribution** describes the **pattern of values** a variable takes
- It shows how frequently different values occur **relative to each other**

For quantitative variables:

- Described using **relative frequencies** or **probabilities** of values

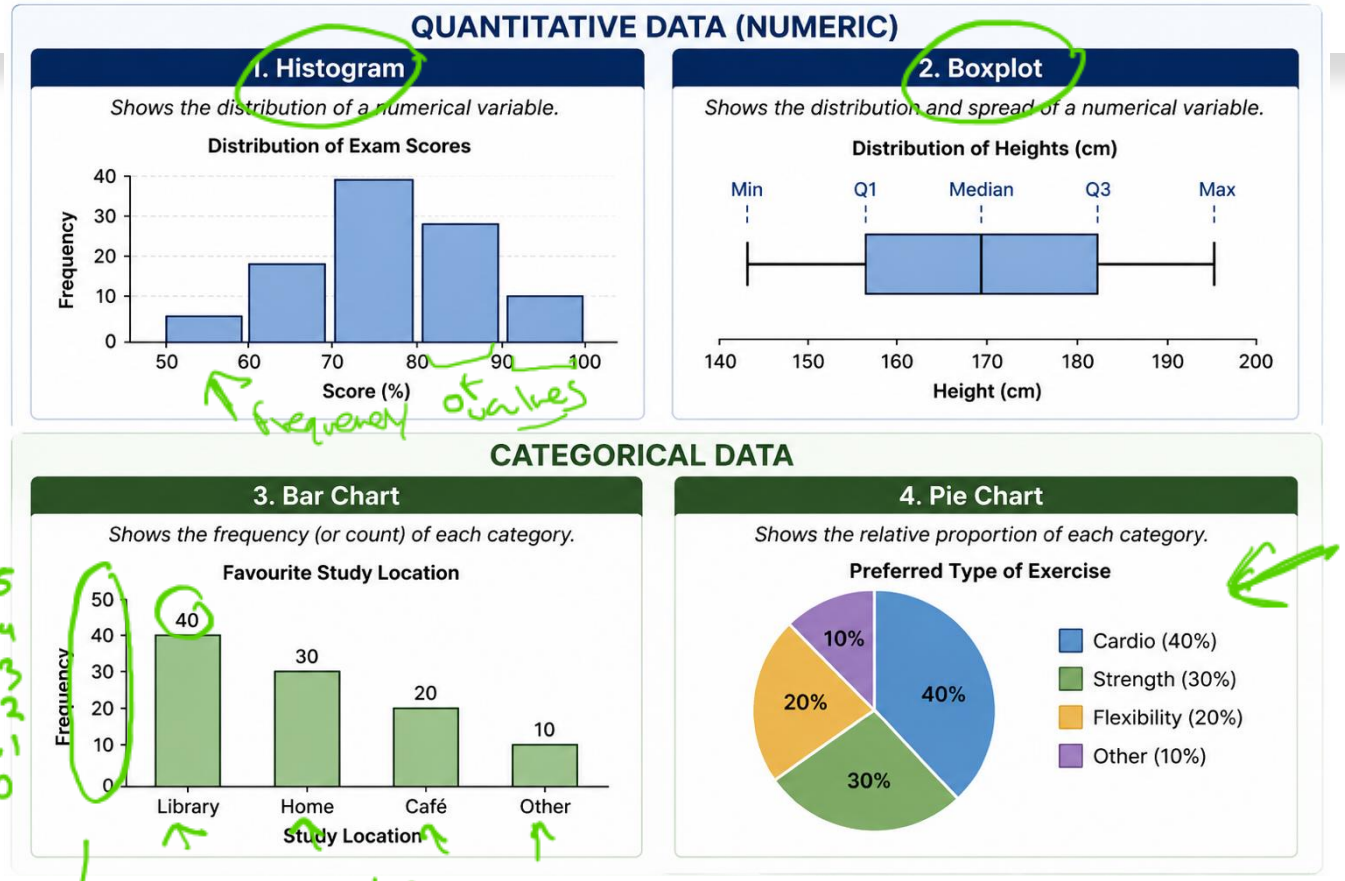
For categorical variables:

- Described using **frequencies** or **relative frequencies** of categories

Beyond One Variable

- We can also study the **joint distribution** of multiple variables

Categorical Variables

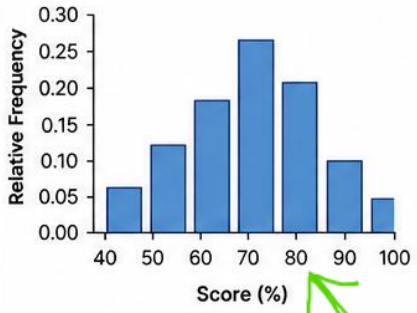


Distributions of Quantitative and Categorical Variables

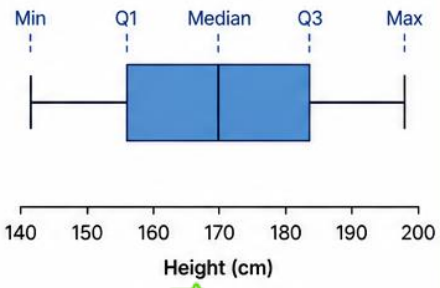
The distribution of a variable is the pattern of values in the data for that variable, showing the frequency of the occurrence of the values relative to each other.

For quantitative variables,
that can be shown by the relative frequencies/probabilities of all possible values of the variable.

Example 1: Histogram
Distribution of Exam Scores

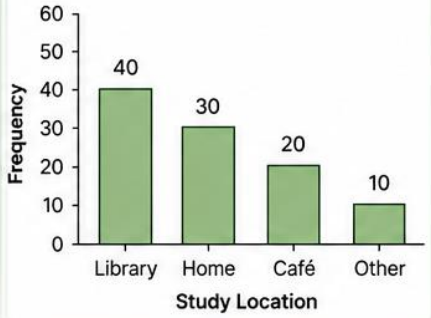


Example 2: Boxplot
Distribution of Heights (cm)

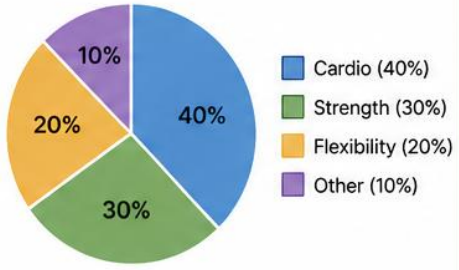


For categorical variables,
that can be shown by frequencies or relative frequencies of each of the categories of the variable.

Example 1: Bar Chart
Favourite Study Location



Example 2: Pie Chart
Preferred Type of Exercise

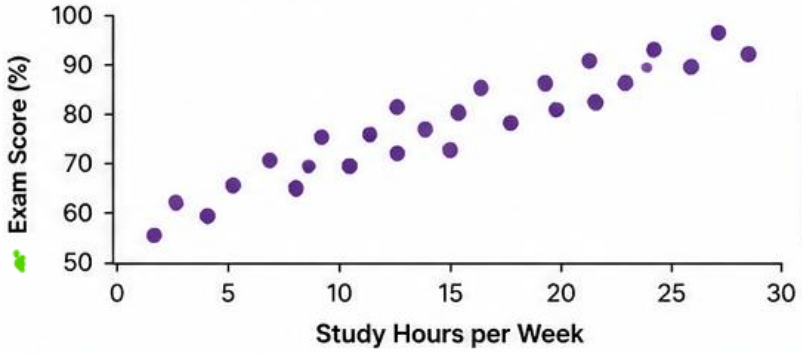


Distributions of Quantitative and Categorical Variables

We can also look at the distribution of multiple variables jointly.

Example 1: Scatter Plot

Relationship between Study Hours and Exam Score



Each point represents one student.

Example 2: Heatmap (Two Categorical Variables)

Relationship between Gender and Favourite Study Location

Gender	Library	Home	Café	Other
Female	30	25	15	10
Male	20	25	15	10

Study Location

Relative Frequency: 0.30, 0.20, 0.10, 0.00

use scatter plots for 2 quantitative vars.

2 categorical variables

Summarizing Variables: Two Variables

Relationship Between Quantitative and Categorical variables

Numerical summaries (5-number summary)

•Compare centre (median) and spread (IQR, range) across categories

•Useful for identifying shifts in distribution and variability

Graphical summaries

•Boxplots or density plots for each category

•Reveal differences in shape, skewness, and outliers

Relationships between two quantitative variables

•Scatterplots to visualize association

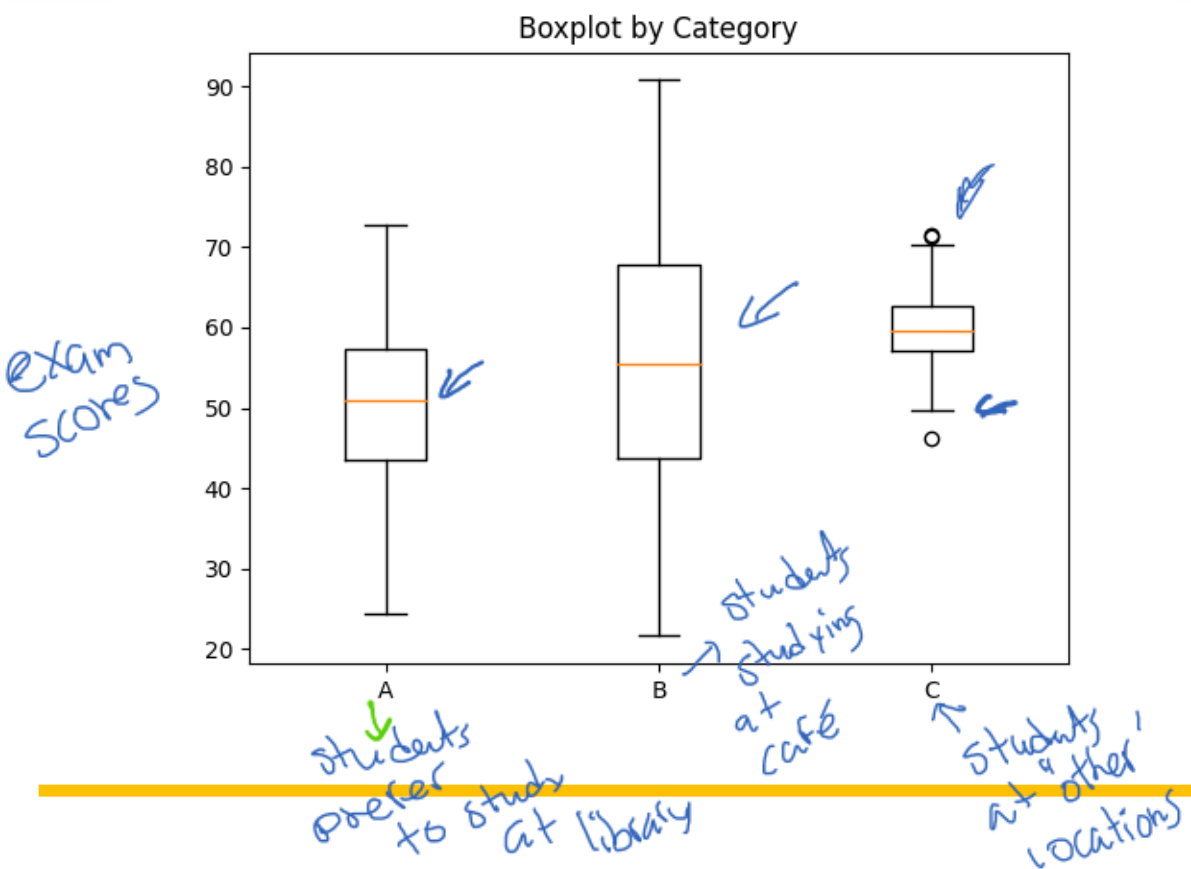
•Correlation to measure strength and direction of linear relationship

Relationships between categorical variables

•Contingency tables to summarize joint frequencies

•Used to assess association between categories (often visualized with stacked bar plots or heatmaps)

Numerical Summaries/Plots



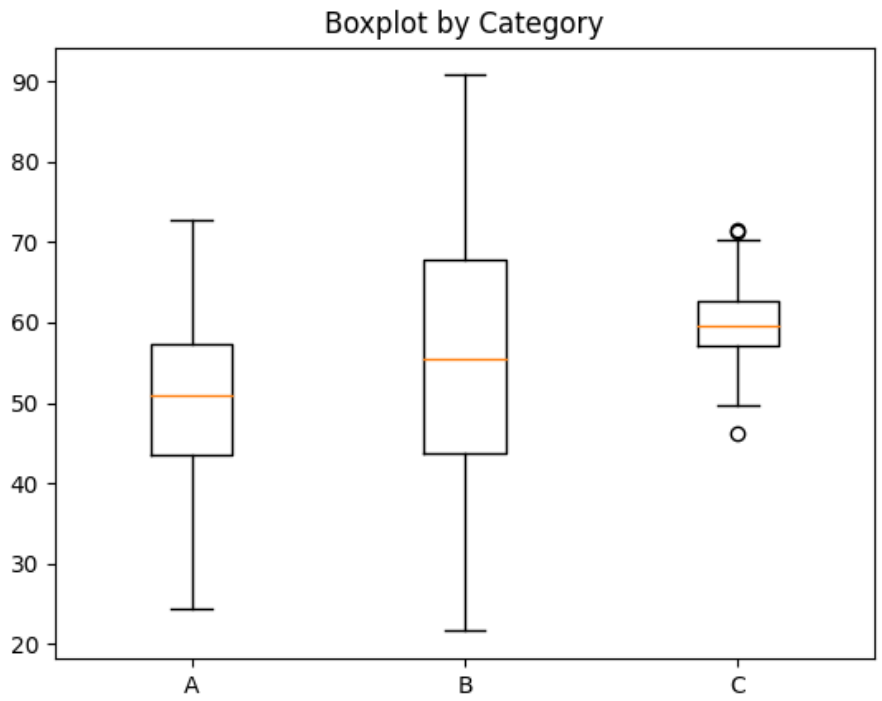
Group A:
Min: 24.47, Q1: 43.56
Median: 50.94, Q3: 57.37
Max: 72.70

Group B:
Min: 21.65, Q1: 43.82
Median: 55.37, Q3: 67.71
Max: 90.75

Group C:
Min: 46.14, Q1: 57.02
Median: 59.62, Q3: 62.69
Max: 71.52

Numerical Summaries/Plots

1 Quantitative
+
1 Qualitative variable



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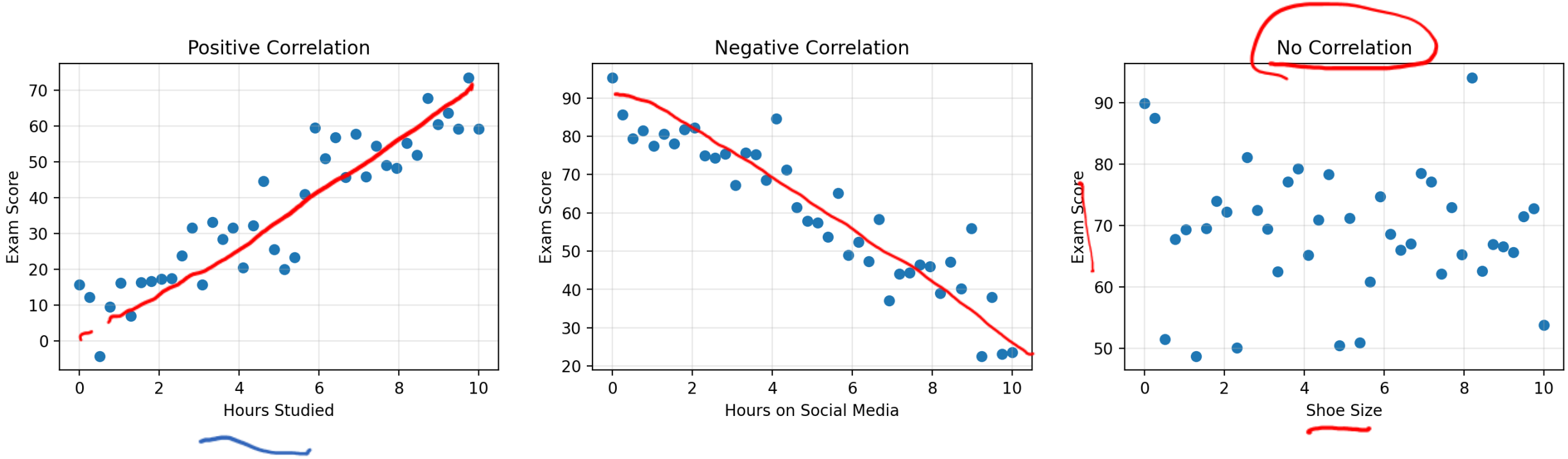
Max: 71.52

Scatterplots → 2 Quantitative variables only!



Contingency Table

2 categorical variables

A common way to study the relationship between two categorical variables is through a **contingency table**.

For example, consider a survey that asks:

- Do you drink coffee? (Yes/No) ← categories
- Do you drink tea? (Yes/No)

We can summarize the joint distribution of responses in a table:

	Tea: Yes	Tea: No	Total
Coffee: Yes	40	30	70
Coffee: No	20	10	30
Total	60	40	100

Contingency Table

Joint Distribution
The **joint distribution** describes how two variables behave *together*.

In a contingency table, this corresponds to the **inside cells**

It shows the frequency (or proportion) for each combination of categories

Example:
Coffee (Yes/No) and Tea (Yes/No) considered simultaneously

	Tea: Yes	Tea: No	Total
Coffee: Yes	40	30	70
Coffee: No	20	10	30
Total	60	40	100

Contingency Table

ignoring tea

⇒ 70 drink coffee

⇒ 30 don't drink coffee

Marginal Distribution
The **marginal distribution** describes each variable *on its own*, ignoring the other variable.
In a contingency table, this corresponds to the **margins (row and column totals)**

It shows the distribution of just one variable

Example:
Overall proportion of coffee drinkers (ignoring tea)

Overall proportion of tea drinkers (ignoring coffee)

ignoring "coffee"

	Tea: Yes	Tea: No	Total
Coffee: Yes	40	30	70
Coffee: No	20	10	30
Total	60	40	100

← marginal distrib of coffee drinkers

⇒ 60 are tea drinkers, 40 are not tea drinkers

Let’s Head Over to R!

Class 1
R Exercise

Lecture Summary

In this lecture, we covered both parts of **Module 1: Summarizing Data**.

- **What to Focus On**
- For now, you can **skip**:
- Resources involving the use of **R**
- The section on **Relationships Between Two Quantitative Variables**

- **R Setup**
- Make sure you know how to get set up using **R**.
- Instructions are provided in the **syllabus**
- If you encounter any issues:
    - Post on **Piazza**
    - Attend **office hours**

- **Practice Problems and Solutions for Week 1** are available on **Quercus**

Looking Ahead

- **Quiz 1 (and bonus AI usage survey) is due on May 11 at 11:59 PM** (opens on May 8 at 12:00 AM)

- Class 2 (Thursday) we will cover probability